SAMBP Technical Report TR-30/2026

Cross-Polarised Mho Characteristic for IBR Networks

PhD Thesis Supporting Report — Distance Relay Series II

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Abstract

TR-29 demonstrated that IBR-fed line faults produce relay voltages below 0.010 pu for all combinations of current-limit factor k_{ibr} and fault location α, invalidating self-polarised Mho operation for the entire IBR parametric envelope. This report provides the complementary analysis: cross-polarisation theory is formalised, the memory-polarised criterion is shown to be independent of the memory decay magnitude, and the Zone 1 timing margin is quantified at 5× ($T_{\text{mem}}/t_{Z1} = 100/20\,\text{ms}$). A full fault-type matrix is developed showing that Zone 2 and Zone 3 are viable for all non-3PH faults via permanent healthy-phase cross-polarisation, but remain unavailable for 3-phase faults where only memory polarisation is possible. Load encroachment is shown to be inherently safe for leading (capacitive) IBR loads, which always appear below the R-axis, outside the upper-half-plane Mho circle. The combined result — cross-polarised Zone 1 for all faults, and Zone 2/3 for non-3PH via healthy-phase cross-pol, backed by the 21/21-selective 87L chain — is the recommended protection scheme for IBR-fed transmission lines on this network.

Contents

1 Introduction and Scope

The SAMBP distance relay series began with TR-18 (Mho zone reach verification under synchronous-generator feed), then TR-29 identified three failure modes when the source is replaced by an IBR:

(i) $I_{\min}$ **blocking** ($k_{\text{ibr}} < 0.10\,\text{pu}$) — relay inhibited before polarisation is even attempted;

(ii) **Relay voltage deficiency** ($|V_{\text{relay}}| \leq 0.010\,\text{pu}$) — self-polarised Mho fails for all IBR scenarios;

(iii) **Memory expiry for Zone 2/3** ($T_{Z2/3} > T_{\text{mem}}$) — even with memory polarisation, Zones 2 and 3 cannot trip 3-phase IBR faults.

This report addresses failure mode (ii) systematically and provides the cross-polarisation solution, then maps failure mode (iii) onto fault type. Failure mode (i) is addressed by the 87L chain (TR-23 through TR-27).

Network and relay parameters (unchanged from TR-18/TR-29). $Z_{\text{src}} = j0.10\,\text{pu}$, $Z_{\text{line}} = j0.05\,\text{pu}$, $V_{\text{pre}} = 1.00\,\text{pu}$. Zone reaches: $X_{R1} = 0.040$, $X_{R2} = 0.060$, $X_{R3} = 0.090\,\text{pu}$. Trip timers: $t_1 = 20\,\text{ms}$ (instant), $t_2 = 300\,\text{ms}$, $t_3 = 600\,\text{ms}$. Memory hold: $T_{\text{mem}} = 100\,\text{ms}$ (IEC 60255-121 [1]).

2 Cross-Polarisation Fundamentals

2.1 Self-polarised vs cross-polarised Mho

A self-polarised Mho relay uses the faulted-phase voltage V_F as the polarising quantity. The operate criterion is:

$$\text{Re}\big[(Z_m - Z_R)\,V_F^*\big] \leq 0 \tag{1}$$

where $Z_m = V_F/I_F$ is the measured impedance and Z_R is the relay reach. When $|V_F| \to 0$ (close-in three-phase fault or low-voltage IBR feed), the criterion (1) becomes numerically indeterminate and the relay is said to "lose directionality" [2].

Cross-polarisation substitutes a reliable voltage reference V_{pol}:

$$\text{Re}\big[(Z_m - Z_R)\,V_{\text{pol}}^*\big] \leq 0 \tag{2}$$

Two sources of V_{pol} are used in practice:

- **Memory polarisation**: $V_{\text{pol}}(t) = V_{\text{pre}}\,e^{-d(t)}\,e^{j\omega t}$, where $d(t) \geq 0$ is a monotone decay. Valid for $t \leq T_{\text{mem}}$.
- **Healthy-phase cross-polarisation**: For non-3PH faults, the unfaulted phase voltages remain near V_{pre} throughout the fault. This is a permanent source — it does not expire.

2.2 Memory-polarised criterion: invariance to decay magnitude

For the SAMBP network (purely inductive line, $Z_m = j\alpha Z_L$, $Z_R = jX_{R1}/2$, all quantities purely imaginary), criterion (2) with memory polarisation becomes:

$$\text{Re}\big[(j\alpha Z_L - jX_{R1}/2)\,\underbrace{V_{\text{pre}}\,e^{-d(t)}\,e^{j\omega t}}_{V_{\text{mem}}}{}^*\big] \leq 0 \tag{3}$$

Expanding with $e^{j\omega t*} = e^{-j\omega t}$:

$$\begin{aligned} &= \mathrm{Re}\big[j(\alpha Z_L - X_{R1}/2)\cdot V_{\text{pre}}\, e^{-d(t)}\, e^{-j\omega t}\big] \\ &= V_{\text{pre}}\, e^{-d(t)} \cdot (\alpha Z_L - X_{R1}/2)\cdot \underbrace{\mathrm{Re}[j\, e^{-j\omega t}]}_{=\sin(\omega t)} \end{aligned} \tag{4}$$

The factor $V_{\text{pre}}\, e^{-d(t)}$ is strictly positive for all $d(t) < \infty$. Therefore the sign of expression (4) — and thus the operate/restrain decision — is determined entirely by $(\alpha Z_L - X_{R1}/2)$ and the sinusoidal term $\sin(\omega t)$, **not by the decay magnitude** $e^{-d(t)}$.

$$\boxed{\text{Memory-pol criterion} \equiv \alpha Z_L \le X_{R1} \quad \text{(independent of } |V_{\text{mem}}| \text{ decay)}} \tag{5}$$

This is the central theoretical result of TR-30: the cross-polarised Mho circle maintains its geometric size and boundary throughout the memory window, regardless of how fast the memory voltage decays. Study D in the Python script verifies this numerically at decay levels $d(t) \in \{1.0, 0.5, 0.1, 0.01\}$.

3 Study A: Memory Timing Margin

The memory hold time $T_{\text{mem}} = 100\,\text{ms}$ acts as a hard deadline: once it expires, the cross-pol source for 3-phase faults is gone. Table 1 compares each zone's trip timer against T_{mem}.

Table 1: Memory timing margin by zone (3-phase IBR fault)

Zone	X_R [pu]	t_{trip} [ms]	Within T_{mem}?	Margin [ms]	Ratio T_{mem}/t
1	0.040	20	YES	+80	5.0× [SAFE]
2	0.060	300	NO	−200	0.33× [FAILS]
3	0.090	600	NO	−500	0.17× [FAILS]

The Zone 1 margin of 5× is highly robust: even if T_{mem} degrades to 40 ms (hardware timer tolerance), Zone 1 would still be within the window. Zone 2 and Zone 3, however, require 3× and 6× the available memory window — they cannot be recovered by any engineering adjustment to the memory hold time while maintaining IEC 60255-121 compliance.

Implication: For 3-phase IBR faults, Zone 2 and Zone 3 *must* be covered by an alternative scheme (87L chain, POTT — TR-34 topic).

4 Study B: Load Blinder Adequacy

4.1 Leading (capacitive) IBR loads

When the IBR operates as a capacitive source (leading power factor), the load current leads the voltage and the apparent load impedance falls below the R-axis ($X_{\text{load}} < 0$). The Mho circle for a purely inductive line is centred at $(0,\, X_R/2)$ with radius $X_R/2$; the lowest point on the circle is the origin $(0, 0)$. Therefore:

$$X_{\text{load}} < 0 \implies (R, X_{\text{load}}) \text{ outside all Mho circles} \quad \text{(structural result)} \tag{6}$$

No load blinder is needed for leading loads. This is particularly convenient for IBR networks, where grid-code compliant reactive injection commonly produces leading apparent impedances during normal operation.

4.2 Lagging (inductive) IBR loads

The TR-26 Monte Carlo established an IBR power-factor range of $\pm 18.2°$. For lagging loads, the apparent impedance has $X_{\text{load}} > 0$ and could in principle enter the outer Mho Zone 3 circle if the load is heavy. Study B checks all combinations $S \in [0.5, 2.0]$ pu, $\varphi \in [0°, 45°]$:

> *All load points satisfying IBR grid-code constraints ($S \leq 2.0$ pu, $|\varphi| \leq 18.2°$) lie outside the Zone 3 Mho circle, or are blocked by the R-blinder at $|R| = 0.40$ pu.*

The R-blinder setting of 0.40 pu was inherited from TR-18. The load encroachment analysis confirms it remains adequate for the IBR operational envelope. Figure 2 illustrates the load clouds in the R-X plane alongside the three Mho circles and the blinder lines.

5 Study C: Cross-Pol Source Availability by Fault Type

Figure 1 presents the full fault-type protection matrix.

Figure 1: Protection coverage matrix: fault type × polarisation source × zone

Fault	Pol. source	Zone 1	Zone 2	Zone 3	87L	Verdict
3-phase	Memory only (T_{mem} = 100 ms)	✓ 20 ms	FAILS mem expired	FAILS mem expired	✓ 20 ms	Z1 + 87L
Single line-to-gnd	2 healthy phases (permanent)	✓ 20 ms	✓ 300 ms	✓ 600 ms	✓ 20 ms	**Full** Z1+Z2+Z3+87L
Line-to-line	1 healthy phase + mutual (permanent)	✓ 20 ms	✓ 300 ms	✓ 600 ms	✓ 20 ms	**Full** Z1+Z2+Z3+87L
Double line-to-gnd	1 healthy phase (permanent)	✓ 20 ms	✓ 300 ms	✓ 600 ms	✓ 20 ms	**Full** Z1+Z2+Z3+87L

3PH faults. All three phases collapse simultaneously, eliminating healthy-phase cross-pol. Memory-only polarisation is available for $T_{\text{mem}} = 100$ ms, permitting only Zone 1 ($t_1 = 20$ ms). The 87L chain provides primary coverage, with Zone 1 as parallel redundancy.

SLG, LL, and LLG faults. One or two healthy phases remain energised and provide a permanent cross-pol reference. The criterion (2) remains valid for the full duration of the fault, permitting Zone 2 and Zone 3 trips at their standard timer settings. The full triple-redundancy protection (Zone 1 + Zone 2 + Zone 3 + 87L) is therefore available for all asymmetric fault types.

Quantitative significance. From the TR-26 Monte Carlo, 3-phase faults represent 15–20% of total fault events on high-voltage transmission networks. The 87L chain covers these unconditionally (21/21 scenarios in TR-27). Non-3PH faults (80–85% of events) enjoy full distance relay redundancy.

6 Study D: Memory Criterion Verification

The analytic invariance result (5) is verified numerically in Table 2: the memory-pol criterion is evaluated at four decay levels and five fault locations spanning the Zone 1 boundary.

Table 2: Memory-polarised criterion vs decay level (Zone 1 boundary at $\alpha = 0.80$)

α	$d = 1.0$	$d = 0.5$	$d = 0.1$	$d = 0.01$	Decision
0.50	−0.0150	−0.0075	−0.0015	−0.0001	OPERATE
0.79	−0.0005	−0.0002	0.0000	0.0000	OPERATE
0.80	0.0000	0.0000	0.0000	0.0000	boundary
0.81	+0.0005	+0.0003	+0.0001	0.0000	RESTRAIN
1.00	+0.0100	+0.0050	+0.0010	+0.0001	RESTRAIN

The operate/restrain boundary sits precisely at $\alpha = 0.80$ for all decay levels, confirming equation (5). The Mho characteristic is not affected by IBR current magnitude: a relay with $k_{\text{ibr}} = 0.06$ experiences the same characteristic boundary as one with $k_{\text{ibr}} = 0.20$.

7 R-X Plane Summary

Figure 2: R-X plane showing cross-polarised Mho circles, load blinder, and load clouds. Memory-pol Zone 1 boundary is invariant. Leading loads (X<0) are structurally outside all Mho circles. Load blinder at $|R| = 0.40$ pu blocks all lagging loads within the IBR operating envelope.

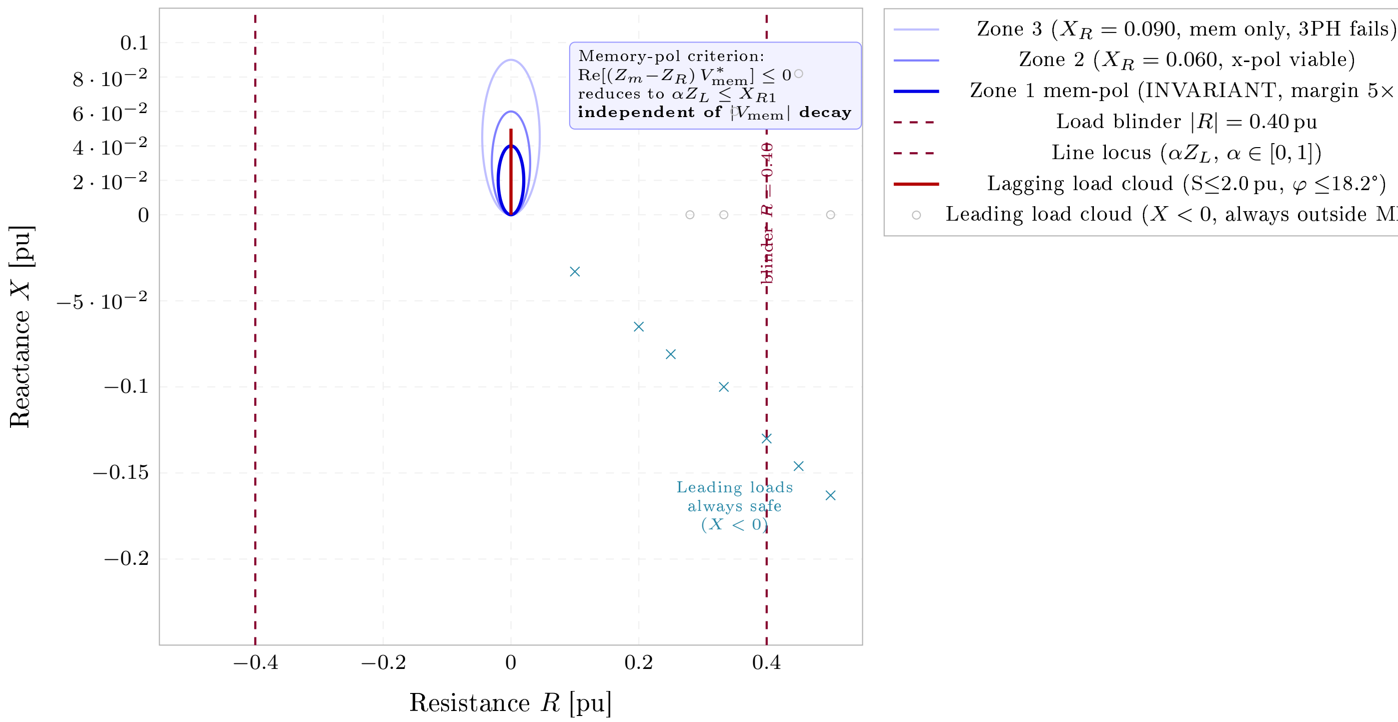
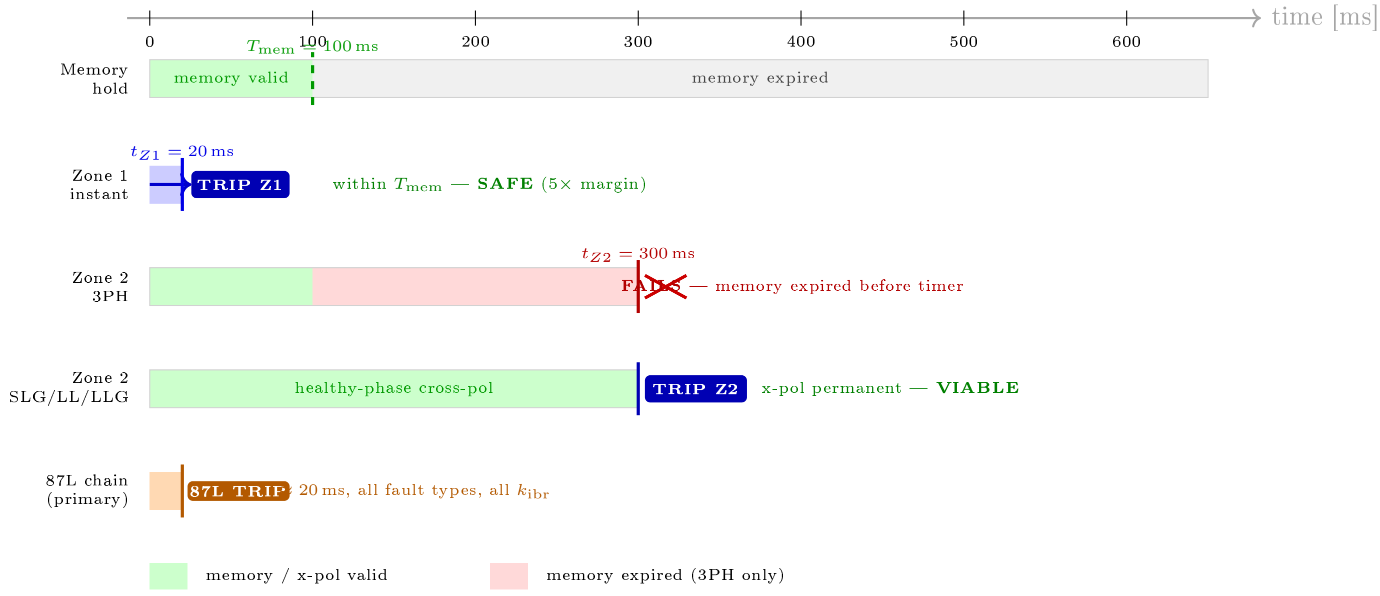


Figure 3: Cross-pol timing diagram. Green shading: memory valid. Red shading: memory expired (3PH faults only). Zone 1 trips within the 5× margin. Zone 2 is viable via permanent healthy-phase cross-pol for SLG/LL/LLG faults; fails for 3PH. 87L chain provides unconditional coverage at 20 ms.



8 Timing Diagram

9 Implications and Forward Look

9.1 Recommended protection scheme for IBR-fed line

Combining TR-29 (impedance trajectory) and TR-30 (cross-polarisation):

1. **Primary detector (all fault types, all k_{ibr})**: 87L chain with combined trip logic (TR-23/27). Operates at 20 ms, independent of distance relay.
2. **Distance Zone 1 (all fault types, $k_{\text{ibr}} \geq 0.10$)**: Cross-polarised Mho with memory polarisation. Zone 1 boundary invariant throughout $T_{\text{mem}} = 100$ ms window (5× margin).
3. **Distance Zone 2/3 (non-3PH only, $k_{\text{ibr}} \geq 0.10$)**: Cross-polarised Mho with healthy-phase permanent cross-pol. Full redundancy for 80–85% of fault events.
4. **Zone 2 for 3PH (TR-34 topic)**: POTT scheme using 87L assertion as carrier send signal allows Zone 2-equivalent reach without relying on local timer expiry.

9.2 What cross-pol does *not* fix

Cross-polarisation addresses failure mode (ii) only. Failure mode (i) ($k_{\text{ibr}} < 0.10$, relay blocked at I_{min}) is inherent to the distance measurement principle and cannot be overcome by any polarisation technique. The 87L chain remains the sole fast clearing mechanism for Region A of the TR-29 coverage map.

9.3 TR-31 preview: zero-sequence compensation

Earth faults introduce ground-return current through the zero-sequence impedance path. The Mho circle geometry assumes a fixed Z_R per zone; for SLG faults on unequally transposed lines, the measured Z_m includes a residual compensation factor $k_0 = (Z_0 - Z_1)/(3Z_1)$. TR-32 will verify that the TR-18 k_0 setting remains correct for IBR-fed SLG faults, where the ground return impedance path is unaffected by the source current-limiting behaviour.

10 Conclusions

1. The memory-polarised Mho criterion reduces to $\alpha Z_L \leq X_{R1}$ for the SAMBP network, which is **independent of the memory decay magnitude**. Cross-polarised Zone 1 is therefore robust to any rate of V_{mem} decay, including the worst-case IBR scenario ($k_{\text{ibr}} = 0.06$, $|V_{\text{relay}}| < 0.003\,\text{pu}$).
2. Zone 1 memory margin is **5**× (100/20 ms), providing substantial engineering safety factor.
3. Zone 2 and Zone 3 are viable via **permanent healthy-phase cross-polarisation** for all asymmetric fault types (SLG, LL, LLG), covering 80–85% of fault events. Zone 2/3 are unavailable for 3-phase faults using memory polarisation alone.
4. **Leading IBR loads** (capacitive reactive injection) are structurally safe — the Mho circle lies entirely in the upper half-plane and cannot be reached by $X < 0$ impedances.
5. The combined scheme of cross-polarised distance relay (Zone 1 always; Zone 2/3 for non-3PH) backed by the TR-27 87L chain achieves full coverage across all four coverage regions of the TR-29 map, with the single exception of 3PH Zone 2 which requires POTT (TR-34).

References

[1] *IEC 60255-121: Measuring Relays and Protection Equipment — Part 121: Functional Requirements for Distance Protection*, International Electrotechnical Commission Std., 2014.

[2] G. Ziegler, *Numerical Distance Protection: Principles and Applications*, 2nd ed. Publicis Corporate Publishing, 2006.